



CareerLaunch

## END-TO-END TEST DOCUMENT

# CareerLaunch

End-to-end acceptance testing —  
every user story, proven against its acceptance criteria.

The formal test document for the CareerLaunch platform: environment and data preparation, forty-six scripted end-to-end test cases across twelve suites (core enrolment engine, automation, console, invoicing, payments, certificates, currency & language, error logging, the student portal, and release quality), a full traceability matrix mapping every acceptance criterion to the cases that prove it, and the sign-off record.

• Core engine & automation

• Payments · certificates · portal

• Traceability & sign-off

### CareerLaunch Platform

Enrolments · Invoicing · Certificates · Payments · Portal

### E2E Test Document · Rev. August 2026

QA & UAT reference · Oxfordable Careers

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# 1 Introduction & test approach

What this document is for, what it covers, and the rules for executing it.

## 1.1 Purpose

This document proves, end to end and against written acceptance criteria, that the CareerLaunch platform behaves as specified — the core enrolment engine (user stories US-01...US-14) and the Extension Pack features (invoicing, payments, certificates, currency & language, error logging, and the Experience Cloud student portal). It is written so that a tester who has never seen the org can execute every case unaided.

## 1.2 References

| DOCUMENT                                                           | USED FOR                                                                       |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Week 2 Participant Workbook v2.0 — Section 1                       | User stories US-01...US-13 and their acceptance criteria; NFR-01...06          |
| Week 3 Participant Workbook — “The three stories we deliver today” | US-10...US-12, US-14 acceptance criteria                                       |
| CareerLaunch Complete Build Guide                                  | Extension feature behaviour (Chapters 7–12) and release procedure (Chapter 14) |
| User Guide / Technical Guide                                       | Expected day-to-day behaviour and architecture reference                       |

## 1.3 Scope

| IN SCOPE                                                                                                                                                                         | OUT OF SCOPE                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All functional acceptance criteria of the core system and Extension Pack, tested through the real UI, real flows, real Apex and the real (stubbed) gateway in a Developer/QA org | Performance/load testing beyond the 200-record bulk case · penetration testing · real payment-provider integration (the gateway runs the deterministic stub) · email deliverability beyond the org's own send |

## 1.4 Entry criteria

- The full application is deployed to the test org and **all unit tests pass** (verified formally again in E2E-45).
- Both permission sets exist and are assigned per Section 2.2.
- Email deliverability is *All Email*; all flows are **Active**; the Experience Cloud site is **published**.
- The seed data of Section 2.3 has been created exactly as written.

## 1.5 Exit criteria

- 100% of applicable cases executed; **every Must-priority case passed**; no open Critical or High defect.
- Suite E (agent) may be marked N/A if Agentforce is unavailable in the org — record N/A, not Fail.
- The traceability matrix (Section 15) shows every acceptance criterion covered by at least one passed case.

## 1.6 Defect recording

Raise every failure with: the test case ID, the step number that failed, expected vs actual, a screenshot, and — where the failure produced one — the **Error Log number** ( ERR-xxxxx from the Error Logs tab), which carries the stack trace the developer needs. Severity scale: **Critical** (money wrong, data loss, security breach) · **High** (acceptance criterion fails) · **Medium** (workaround exists) · **Low** (cosmetic).

## 1.7 How to execute a case

- Run cases **in order within a suite** — later cases reuse earlier state (e.g. G-suite pays the invoice F-suite created).
- Perform every step exactly; do not substitute data. Where a step says “as *Staff Tester*”, log in as that user (Section 2.2).
- Tick each expected result individually. A case passes only when **all** its expected results hold.
- Record Pass / Fail / N/A, tester initials and date in the result strip of each case.

#### RESET RULE

If you must re-run a suite, create fresh records rather than editing used ones — auto-numbers and error logs are audit evidence. Never delete Error Log records during the test cycle.

## 1.8 What to test — at a glance

The complete test surface on one page. Each row is a suite; run them top to bottom. The detailed scripts start in Section 3.

| SUITE                                 | WHAT YOU ARE TESTING                                                                                                                               | HOW                         | CASES       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|
| <b>A — Catalogue &amp; data model</b> | Courses and cohorts live in one place; seat counters compute themselves (and release on cancellation); date validation blocks bad cohorts          | Staff UI                    | E2E-01...04 |
| <b>B — Enrolment rules</b>            | Defaults on create; the capacity rule (one record <i>and</i> a 200-record bulk load); duplicate prevention; the fee-cap validation                 | Staff UI + one Apex script  | E2E-05...09 |
| <b>C — Automation</b>                 | Welcome email + task on confirmation; cohort auto-Full; the Quick Enrol wizard (happy and full paths); the T-3 reminder; the Contact roll-up count | Staff UI + flow debug       | E2E-10...15 |
| <b>D — Enrolment Console</b>          | The staff one-screen console: list/filter with live seat counts, enrol with toast + reactive refresh, friendly error messages                      | Console UI                  | E2E-16...18 |
| <b>E — Agent</b><br>(conditional)     | Availability and status questions answered from live org data; graceful out-of-scope handling                                                      | Agent preview               | E2E-19...21 |
| <b>F — Invoicing</b>                  | Invoices raise themselves on confirmation (fee, Sent, +14 days); never duplicate; skip free courses; Paid demands a reference; cancel → re-issue   | Staff UI                    | E2E-22...26 |
| <b>G — Payments</b>                   | The gateway loop: successful settlement with reference + roll-up; simulated decline changes nothing; double-pay impossible; audit trail complete   | Execute<br>Anonymous        | E2E-27...30 |
| <b>H — Certificates</b>               | Completion produces the PDF, filed on the record, flag ticked, exactly once, with correct content                                                  | Staff UI                    | E2E-31...33 |
| <b>I — Currency &amp; language</b>    | Conversions follow configured rates; rates are frozen on issued invoices; French users get a French portal and certificate                         | Setup + portal              | E2E-34...36 |
| <b>J — Error logging</b>              | Apex exceptions, flow faults and integration errors all become permanent log records — including when the failing transaction rolls back           | Execute<br>Anonymous + flow | E2E-37...39 |
| <b>K — Student portal</b>             | The full customer journey: login, own-records-only privacy, Pay now end to end, certificate download — plus guest and cross-student access control | Portal, two students        | E2E-40...44 |
| <b>L — Release quality</b>            | The automated suite passes at ≥75% coverage; the release validates against a clean org                                                             | CLI                         | E2E-45...46 |

## 2 Environment, roles & test data preparation

Complete this chapter once, before Suite A. Every case assumes this exact starting state.

### 2.1 Environment record

| ITEM                               | VALUE (FILL IN)                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------|
| Org URL / alias                    | _____                                                                                    |
| Org type                           | Developer Edition <input type="checkbox"/> Sandbox <input type="checkbox"/> Other: _____ |
| Deployment commit / tag under test | _____                                                                                    |
| Test cycle start date              | _____                                                                                    |
| Agentforce available in this org?  | Yes <input type="checkbox"/> No <input type="checkbox"/> (if No, Suite E = N/A)          |

### 2.2 Test users (three roles)

| ALIAS IN THIS DOCUMENT       | USER                                                                                                                    | ACCESS                                                                                                                                                                  |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Admin Tester</b>          | System Administrator profile                                                                                            | Both permission sets assigned<br>( <code>Enrolment_Console_User</code> + <code>Student_Portal_User</code> )<br>— required for field-level security after a fresh deploy |
| <b>Staff Tester</b>          | A standard internal user (or the admin acting as staff)                                                                 | <code>Enrolment_Console_User</code> permission set                                                                                                                      |
| <b>Student A / Student B</b> | Two Experience Cloud users (Customer Community User profile), created via <b>Enable Customer User</b> on their Contacts | <code>Student_Portal_User</code> permission set each · known passwords · site membership per Build Guide §12.4                                                          |

### 2.3 Seed data script (TD-01 ... TD-08)

#### AUTOMATED SEEDING AVAILABLE

Running `sf apex run -o <org> -f scripts/apex/e2e_seed.apex` creates TD-01...TD-06 automatically and is idempotent (safe to re-run). The E2E-07 bulk load is likewise scripted as `scripts/apex/e2e_bulk.apex`. In the reference QA org this data — including the TD-07 portal users `e2e.student.a@careerlaunch.portal` and `e2e.student.b@careerlaunch.portal` — is already provisioned; ask the instructor for the portal password. The table below remains the manual, click-by-click equivalent.

| ID    | CREATE (AS ADMIN TESTER) | VALUES                                                                                                         |
|-------|--------------------------|----------------------------------------------------------------------------------------------------------------|
| TD-01 | Course                   | <code>E2E_Developer_Course</code> · Level Introductory · 5 weeks · <b>Fee 500</b> · Active                     |
| TD-02 | Course                   | <code>E2E_Free_Taster</code> · Level Introductory · 1 week · <b>Fee 0</b> · Active                             |
| TD-03 | Cohort                   | On TD-01 · starts today + 14 · ends today + 49 · Online · <b>Capacity 3</b> · Status Open (“the small cohort”) |

|       |                    |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TD-04 | Cohort             | On TD-01 · starts today + 30 · Online · Capacity 200 · Status Open (“the big cohort”)                                                                                                                                                                                                                                                                                                             |
| TD-05 | Cohort             | On TD-02 · starts today + 14 · Online · Capacity 10 · Status Open (“the free cohort”)                                                                                                                                                                                                                                                                                                             |
| TD-06 | Account + Contacts | Account <code>E2E Students</code> ; under it Contacts <b>Student A</b> ( <code>e2e.student.a@oxfordable.test</code> ), <b>Student B</b> ( <code>e2e.student.b@oxfordable.test</code> ), <b>Student C</b> ( <code>e2e.declined@oxfordable.test</code> — the word “declined” in the email is deliberate: it drives the gateway stub's failure path), plus <b>Students D and E</b> (any test emails) |
| TD-07 | Portal users       | Enable Customer User for <b>Student A</b> and <b>Student B</b> (Build Guide §12.5): Customer Community User profile, unique usernames, your inbox as Email, passwords set, permission set assigned                                                                                                                                                                                                |
| TD-08 | Nothing else       | Do <b>not</b> pre-create Enrolments or Invoices — the suites create them so the automation is what gets tested                                                                                                                                                                                                                                                                                    |

#### PREPARATION CHECKPOINT

Tabs visible: Courses, Cohorts, Enrolments, Invoices, Error Logs. TD-03 shows Seats Remaining = 3. Both portal users can reach the site login page. Error Logs tab is empty (or note existing count: \_\_\_\_\_ — later suites assert on *new* records).

### 3 Suite A — Catalogue & data model

Proves US-01 (catalogue in one place), US-03 (seat visibility with no manual entry) and the data-quality validation rules. Run as **Staff Tester** unless stated.

#### E2E-01 Course and cohort creation

US-01 · Must

**PRECONDITIONS** Seed data TD-01...TD-05 exists.

##### STEPS

1. Open the **Courses** tab and open *E2E Developer Course*.
2. Check its **Cohorts** related list.
3. Open the small cohort (TD-03).

##### EXPECTED

- ✓ The course record shows level, duration, fee and active flag.
- ✓ The related list shows both cohorts of TD-01 — the catalogue lives in one place.
- ✓ The cohort shows its Course, start date, delivery mode, capacity and status, and an auto-number like `C0-0001`.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

#### E2E-02 Cohort date validation

US-01 · VR End\_Date\_After\_Start\_Date · Must

##### STEPS

1. Edit the small cohort: set End Date to a date **before** its Start Date → Save.
2. Restore the correct End Date → Save.

##### EXPECTED

- ✓ Step 1 is blocked with “End Date cannot be earlier than Start Date.”
- ✓ Step 2 saves normally.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

#### E2E-03 Seat counters compute themselves

US-03 AC: “both without manual entry” · Must

##### STEPS

1. On the small cohort note Enrolled Count and Seats Remaining (expect 0 / 3).
2. Create an Enrolment: Cohort = small cohort, Student = **Student D**, leave Status blank → Save.
3. Refresh the cohort.

##### EXPECTED

- ✓ Enrolled Count = 1, Seats Remaining = 2 — no field was typed by hand.
- ✓ The Cohort Label formula reads “E2E Developer Course - <start date>”.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## STEPS

1. Edit Student D's enrolment from E2E-03: Status = **Cancelled** → Save.
2. Refresh the small cohort.
3. Set the enrolment back to **Applied** → Save (it is reused later).

## EXPECTED

- ✓ After step 1: Enrolled Count = 0, Seats Remaining = 3 (roll-up filter excludes Cancelled).
- ✓ After step 3: counts return to 1 / 2.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_



## 4 Suite B — Enrolment business rules

Proves US-02 (defaults), US-04 (capacity, including the 200-record bulk criterion NFR-01), US-05 (duplicates) and the payment validation rule. Run as **Staff Tester**; E2E-07 needs Admin Tester.

### E2E-05 Defaults applied on create

US-02 AC: date defaults today · status defaults Applied · Must

#### STEPS

1. Create an Enrolment: Cohort = small cohort, Student = **Student E**; leave Status and Enrolment Date **blank** → Save.
2. Open the saved record.

#### EXPECTED

- ✓ Status = **Applied** · Enrolment Date = **today** · Amount Paid = **0**.
- ✓ Enrolment Key is populated in the form *CohortId-StudentId*.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-06 Capacity enforced — single record

US-04 AC: “fails with a clear error” · Must

#### PRECONDITIONS

Small cohort (capacity 3) holds 2 live enrolments (Students D and E). Add a third — **Student A**, Status Applied — so the cohort is full.

#### STEPS

1. Attempt a fourth Enrolment on the small cohort with **Student B** → Save.

#### EXPECTED

- ✓ The save is blocked with the message “*This cohort is full. Please choose another cohort or add the student to the waiting list.*” — a sentence, not a stack trace (NFR-04).
- ✓ The cohort still shows exactly 3 enrolments.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-07 Capacity holds under a 200-record bulk load

US-04 AC: “the rule holds when 200 records are loaded at once” · NFR-01 · Must

#### PRECONDITIONS

As **Admin Tester**. The big cohort (TD-04, capacity 200) has 0 enrolments. This case runs the platform's own bulk path via Execute Anonymous — equivalent to a Data Loader insert.

#### STEPS

1. Developer Console → Debug → Execute Anonymous. Run a script that: creates 205 test Contacts, builds 205 Enrolments against the big cohort (Status Confirmed), and inserts them with `Database.insert(list, false)`, then debugs the success count. (Your instructor's seed script `scripts/apex/e2e_bulk.apex` contains this verbatim.)
2. Open the big cohort.

#### EXPECTED

- ✓ Exactly **200** inserts succeed; **5** fail with the capacity message.
- ✓ Cohort: Enrolled Count = 200, Seats Remaining = 0 — and no `Too many SOQL queries` error occurred (bulk-safe).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

**E2E-08 Duplicate enrolment rejected**

**US-05 AC: “a second enrolment for the same Contact and Cohort is rejected” · Must**

**STEPS**

1. Attempt a second Enrolment for **Student E** on the small cohort → Save.

**EXPECTED**

✓ The save is rejected (duplicate/unique-key error referencing the Enrolment Key).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

**E2E-09 Payment cannot exceed the fee**

**VR Payment\_Not\_Above\_Fee (cross-object) · Must**

**STEPS**

1. Edit Student E's enrolment: Amount Paid = **600** (fee is 500) → Save.
2. Set Amount Paid back to 0 → Save.

**EXPECTED**

- ✓ Step 1 blocked: “Amount Paid cannot exceed the course fee.”
- ✓ Step 2 saves.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 5 Suite C — Automation

Proves US-06, US-07, US-08, US-09 and US-13. Run as **Staff Tester**. Note: confirming an enrolment also raises an invoice (Suite F asserts on those — do not delete them).

### E2E-10 Confirmation sends the welcome email and creates the task

US-06 both ACs · Must

#### STEPS

1. Edit **Student D's** enrolment on the small cohort: Status = **Confirmed** → Save.
2. Check the enrolment's Activity/Tasks area, and the inbox of Student D's email (your test inbox).

#### EXPECTED

- ✓ A Task exists: subject “Send welcome pack to Student D”, due today, priority High, owned by the record owner.
- ✓ The welcome email arrived with the cohort label in the subject.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-11 Cohort flips itself to Full on the last seat

US-07 AC: “without manual intervention” · Must

#### STEPS

1. Confirm the remaining two enrolments on the small cohort (Students E and A), one at a time.
2. Refresh the cohort after the last confirmation.

#### EXPECTED

- ✓ When Seats Remaining reaches 0, Cohort Status = **Full** — nobody edited the cohort.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-12 Quick Enrol wizard — happy path

US-08 AC part 1 · Should

#### STEPS

1. Launch **Quick Enrol Student** (App Launcher or the Quick Enrol button).
2. Course = E2E Developer Course → cohort = **the big cohort** → student = **Student B** → continue to the end.

#### EXPECTED

- ✓ The confirmation screen shows the new enrolment number.
- ✓ The enrolment exists with Status Confirmed.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-13 Quick Enrol wizard — full cohort path

US-08 AC part 2: “showing an error if the cohort is full” · Should

#### STEPS

1. Run the wizard again and choose the (now Full) **small cohort**, any remaining student.

#### EXPECTED

- ✓ Either the full cohort is not offered (Status ≠ Open) or the “that cohort just filled up” screen appears — no enrolment is created either way, and no raw fault text is shown.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

**E2E-14 Pre-start reminder targets the right records** **US-09 AC: “three days before Start Date, for confirmed enrolments only” · Should**

**PRECONDITIONS** As Admin Tester. Create a throwaway cohort on TD-01 starting **today + 3**, capacity 5, Open; add one **Confirmed** and one **Applied** enrolment (any spare test contacts).

**STEPS**

1. Setup → Flows → `Cohort_Start_Reminder` → open the latest version → **Debug** → Run (or wait for the 07:00 scheduled run).
2. Check Tasks/emails for both enrolments.

**EXPECTED** ✓ The **Confirmed** enrolment received the reminder; the **Applied** one did not.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

**E2E-15 Contact shows its live enrolment count** **US-13 AC: “updated automatically after enrolment changes” · Could**

**STEPS**

1. Open **Student B's** Contact record (they hold one enrolment from E2E-12) and note **Total Enrolments**.
2. Cancel that enrolment (Status = Cancelled), wait a few seconds, refresh the Contact.
3. Restore it to Confirmed, refresh again.

**EXPECTED** ✓ Count reads 1 → drops to 0 after cancellation → returns to 1 after restore (asynchronous — allow a few seconds each time).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 6 Suite D — Enrolment Console

Proves US-10 and NFR-04 through the staff UI. Run as **Staff Tester**.

### E2E-16 Console lists and filters open cohorts

US-10 AC part 1 · Must

#### STEPS

1. App Launcher → **Enrolment Console**.
2. Observe the Open Cohorts table, then filter by course = E2E Developer Course, then back to All courses.

#### EXPECTED

- ✓ Only **Open**, future-dated cohorts appear, each showing **seats remaining**; the Full small cohort is absent.
- ✓ Filtering narrows and restores the list reactively, with no page reload.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-17 Enrol from the console — toast, refresh, navigation

US-10 AC part 2 · Must

#### STEPS

1. Select the **big cohort** row; in the Enrol panel pick **Student D** → **Enrol student**.

#### EXPECTED

- ✓ A success toast shows the new enrolment number.
- ✓ The table's seat count for that cohort **drops by one** without a manual refresh.
- ✓ The browser navigates to the new enrolment record.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-18 Console errors are sentences, not stack traces

NFR-04 · US-04/05 through the UI · Must

#### STEPS

1. In the console, attempt to enrol **Student D** onto the big cohort **again** (duplicate).

#### EXPECTED

- ✓ An error toast reads *"That student is already enrolled on this cohort."* — no ids, no exception class names, no stack trace.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 7 Suite E — Agentforce agent (conditional)

Proves US-11 and US-12. If Agentforce is unavailable in this org, mark all three cases N/A (§2.1) — the invocable Apex they exercise is already covered by unit tests in Suite L.

### E2E-19 Availability question answered from live data

US-11 AC · Must\*

#### STEPS

1. Open the agent's conversation preview and ask: *"Do you have any Developer cohorts with space?"*

#### EXPECTED

- ✓ The reply lists real open cohorts with start dates and seats remaining that match the Cohorts tab — nothing invented.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-20 Enrolment status by email

US-12 AC · Must\*

#### STEPS

1. Ask: *"What is the status of my enrolment?"* — the agent should ask for the booking email.
2. Reply with `e2e.student.a@oxfordable.test`.

#### EXPECTED

- ✓ The reply names Student A's cohort, start date and status, matching the org.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-21 Out-of-scope question declined gracefully

Agent topic scope · Should

#### STEPS

1. Ask: *"How much does parking cost at your office?"*

#### EXPECTED

- ✓ A polite out-of-scope answer offering to pass it to the team; nothing invented.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 8 Suite F — Invoicing

Acceptance criteria: one Sent invoice per confirmed enrolment at the course fee, due in 14 days; re-saves never duplicate; free courses produce nothing; Paid requires a gateway reference; cancelling allows a re-issue. Run as **Staff Tester**.

### E2E-22 Invoice raises itself on confirmation

EXT-INV-1 · Must

**PRECONDITIONS** Student D's big-cohort enrolment from E2E-17 exists at Status Confirmed (if E2E-17 created it as Confirmed, its invoice already exists — use it).

**STEPS** 1. Open that enrolment's **Invoices** related list and open the invoice.

**EXPECTED** ✓ Exactly one invoice: auto-number INV-xxxxx · Status **Sent** · Amount (GBP) = **500** (the fee) · Currency GBP · Exchange Rate 1.000000 · Amount (Invoice Currency) 500.00 · Due Date = **creation date + 14**.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-23 Re-saving a confirmed enrolment never duplicates the invoice

EXT-INV-2 (idempotency) · Must

**STEPS** 1. Edit the same enrolment: set Attendance % = 10 → Save. Repeat with 20 → Save.  
2. Re-open the Invoices related list.

**EXPECTED** ✓ Still exactly **one** invoice.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-24 Free courses are never invoiced

EXT-INV-3 · Must

**STEPS** 1. Create an Enrolment on the **free cohort** (TD-05, fee 0): Student = any spare contact, Status = **Confirmed** → Save.  
2. Check its Invoices related list.

**EXPECTED** ✓ No invoice exists.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-25 Paid requires a gateway reference

VR Paid\_Requires\_Reference · Must

**STEPS** 1. Edit Student D's invoice: Status = **Paid**, leave Payment Reference blank → Save.

**EXPECTED** ✓ Blocked: “An invoice cannot be marked Paid without the gateway payment reference.” The invoice remains Sent.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## E2E-26 Cancelling an invoice allows a fresh one

EXT-INV-4 · Should

### STEPS

1. Edit Student D's invoice: Status = **Cancelled** → Save.
2. Edit the enrolment (any field, e.g. Attendance % = 30) → Save.
3. Check the Invoices related list.

### EXPECTED

✓ A **new** Sent invoice exists alongside the Cancelled one (cancelled invoices don't count as live).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_



## 9 Suite G — Payments

Acceptance criteria: a successful charge marks the invoice Paid with the gateway reference and date, and rolls the amount onto the enrolment; a declined card changes nothing; a paid invoice can never be paid again. The org runs the deterministic stub — an email containing “declined” simulates a failed card. Run as **Admin Tester** (Execute Anonymous); the portal path is re-proven in Suite K.

### E2E-27 Successful payment settles the invoice

EXT-PAY-1 · Must

#### STEPS

1. Copy the Id of Student D's **new Sent invoice** (from E2E-26) out of its URL.

2. Execute Anonymous:

```
System.debug(JSON.serialize(PaymentService.payInvoice('<invoiceId>')));
```

3. Refresh the invoice and its enrolment.

#### EXPECTED

✓ Debug shows `"success":true` and a transaction id of the form **STUB-INV-xxxxx-GBP**.

✓ Invoice: Status **Paid** · Paid Date = today · Payment Reference = that STUB id.

✓ Enrolment: **Amount Paid = 500** (and the Suite B fee-cap rule accepted it).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-28 Declined card leaves everything untouched

EXT-PAY-2 · Must

**PRECONDITIONS** Enrol **Student C** ( `e2e.declined@...` ) on the big cohort, Status Confirmed → their invoice auto-raises.

#### STEPS

1. Pay Student C's invoice via Execute Anonymous as in E2E-27.

2. Refresh the invoice and enrolment.

#### EXPECTED

✓ Debug shows `"success":false` with a simulated-decline message.

✓ Invoice remains **Sent**, no reference, no paid date; enrolment Amount Paid remains 0.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-29 A paid invoice cannot be paid twice

EXT-PAY-3 (idempotent settlement) · Must

#### STEPS

1. Re-run the E2E-27 payment against the **same, now Paid** invoice id.

#### EXPECTED

✓ A handled exception stating the invoice is **already paid**; Amount Paid on the enrolment is unchanged (still 500, not 1000).

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

STEPS

- 1. On Student D's paid invoice, verify every audit field in one pass.
- 2. Open the **Error Logs** tab.

EXPECTED

- ✔ Status Paid + Paid Date + Payment Reference are all populated and mutually consistent.
- ✔ No new Error Log rows were produced by the successful payment path.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 10 Suite H — Certificates

Acceptance criteria: completing an enrolment produces a PDF certificate filed on the record with the flag ticked — exactly once — and the PDF carries the student's real details. Run as **Staff Tester**.

### E2E-31 Certificate issues on completion

EXT-CERT-1 · Must

#### STEPS

1. Edit **Student D's** big-cohort enrolment: Status = **Completed** → Save.
2. Wait ~10 seconds (the render is asynchronous), refresh the record.

#### EXPECTED

- ✓ **Certificate Issued** is ticked.
- ✓ The Files related list holds `Certificate ENR-xxxxx`.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-32 A certificate is only ever issued once

EXT-CERT-2 · Must

#### STEPS

1. Set the same enrolment's Status to Confirmed → Save, then back to **Completed** → Save.
2. Wait, refresh, count the files.

#### EXPECTED

- ✓ Still exactly **one** certificate file.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-33 The PDF content is correct

EXT-CERT-3 · Must

#### STEPS

1. Open the certificate file and inspect it.

#### EXPECTED

- ✓ A real PDF (A4 landscape, designed layout), showing: the student's **name**, the **course name and level**, the cohort label, the issue date, and the enrolment number — all matching the records.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 11 Suite I — Currency & language

Acceptance criteria: conversions follow the configured rates; the rate is frozen on each invoice at issue time; the UI and certificate follow the user's language. E2E-34/35 as **Admin Tester**; E2E-36 uses Student A.

### E2E-34 Conversion follows the configured rate

EXT-CCY-1 · Must

#### STEPS

1. Setup → Custom Metadata Types → Exchange Rate → Manage Records → note the USD rate (default 1.27).
2. Execute Anonymous: `System.debug(CurrencyService.convertFromGBP(100, 'USD'));` and `System.debug(CurrencyService.format(1250, 'GBP'));`

#### EXPECTED

- ✓ First debug =  $100 \times$  the USD rate, to 2 dp (127.00 at default) · second = **£1,250** formatted with symbol and separator.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-35 Rates are frozen on existing invoices

EXT-CCY-2 · Must

#### STEPS

1. Note Amount (Invoice Currency) on Student D's **Paid** invoice.
2. Edit the **GBP** Exchange Rate record: Rate From GBP = 2.0 → Save.
3. Refresh the invoice, then create a brand-new Confirmed enrolment (any spare student, big cohort) and open its fresh invoice.
4. **Restore the GBP rate to 1.0** and cancel/clean the throwaway enrolment.

#### EXPECTED

- ✓ The existing invoice is **unchanged** (its stored rate stays 1.000000).
- ✓ The new invoice was stamped with the **new** rate (2.000000 · Amount In Currency = 1000.00) — future invoices only.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-36 French user sees a French application

EXT-I18N-1 · Must

#### STEPS

1. Setup → Users → **Student A's portal user** → Edit → Language = `Français` → Save.
2. Log in to the portal as Student A (private window) and read the two cards.
3. Restore the language to English afterwards.

#### EXPECTED

- ✓ Cards read **Mes inscriptions / Mes factures**; a payable invoice's button reads **Payer maintenant**.
- ✓ (If a certificate is issued while French is active, its headings render in French too.)

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 12 Suite J — Error logging

Acceptance criteria: every failure — Apex, flow fault, integration — becomes a permanent Error Log record with source, message, stack trace and severity, even when the failing transaction rolls back. Run as **Admin Tester**.

### E2E-37 Apex exceptions are logged

EXT-ERR-1 · Must

#### STEPS

1. Execute Anonymous:

```
try { Integer boom = 1/0; } catch (Exception e) { ErrorLogger.log(e, 'E2E-37'); }
```

2. Open the Error Logs tab.

#### EXPECTED

✓ A new **ERR-** record: Source Type **Apex**, Source Name **E2E-37**, message containing “Divide by 0”, a populated stack trace, Severity Error.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-38 Flow faults are logged

EXT-ERR-2 · Must

#### STEPS

1. Force a fault in **Quick\_Enrol\_Student** — e.g. run it against the Full small cohort so the Create element fails (or temporarily deactivate a referenced field per your instructor).
2. Check the Error Logs tab.

#### EXPECTED

✓ A new record: Source Type **Flow**, Source Name = the flow's name, message = the platform fault text.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-39 Logs survive a transaction rollback

EXT-ERR-3 (the whole point of the design) · Must

#### STEPS

1. Execute Anonymous — note this logs, **then deliberately crashes the same transaction:**

```
ErrorLogger.log('Apex','E2E-39','before the crash',null,null,'Critical');  
insert new Course__c(); // missing required Name → the whole transaction rolls  
back
```

2. Confirm the Execute Anonymous run **failed** (that is the test).
3. Open the Error Logs tab.

#### EXPECTED

✓ The **E2E-39** record exists with Severity **Critical** — the platform event escaped the rollback that destroyed everything else in that transaction.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 13 Suite K — Student portal, end to end

The customer journey plus its security boundaries. Uses **Student A** and **Student B** in private/incognito windows. Preconditions: Student A holds a Confirmed enrolment with a Sent invoice; Student B holds at least one enrolment (from earlier suites — create any missing state as Staff Tester first).

### E2E-40 Portal login and dashboard

EXT-POR-1 · Must

#### STEPS

1. Private window → site URL + `/login` → sign in as **Student A**.

#### EXPECTED

- ✓ Login succeeds; **My Enrolments** shows their course, cohort, start date and status badge; **My Invoices** shows their invoice with amount, currency, due date and status.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-41 A student sees only their own records

EXT-POR-2 (privacy) · Must

#### STEPS

1. Still as Student A: count the rows in both cards.
2. In a second private window, log in as **Student B** and compare.

#### EXPECTED

- ✓ Student A sees only A's records; Student B sees only B's. Neither sees the other's enrolments or invoices anywhere.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-42 Pay now, end to end

EXT-POR-3 + EXT-PAY-1 through the UI · Must

#### STEPS

1. As Student A, press **Pay now** on the Sent invoice.
2. As Admin Tester, verify the invoice and enrolment records.

#### EXPECTED

- ✓ Toast *"Payment received - thank you!"* with a STUB reference; the badge flips to **Paid**; the button disappears; the list refreshes without a manual reload.
- ✓ Back end: Status Paid, reference + date stored, enrolment Amount Paid = fee, no new Error Logs.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-43 Certificate download from the portal

EXT-POR-4 · Must

#### STEPS

1. As Staff Tester, set Student A's enrolment to **Completed**; wait for the certificate (Suite H behaviour).
2. As Student A, refresh the portal.

#### EXPECTED

- ✓ A **Download certificate** link appears under the completed course and downloads the correct PDF.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## STEPS

1. In a fresh private window, open a portal page URL **without logging in**.
2. Cross-student server check: this is proven by the automated test `StudentPortalControllerTest.payInvoiceRejectsSomeoneElsesInvoice` — confirm it passed in the E2E-45 run and record its name here.

## EXPECTED

- ✓ The guest is redirected to login (or shown no data) — never records.
- ✓ The named unit test is green: the server rejects paying another student's invoice even with a valid record id.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

## 14 Suite L — Release quality

US-14: covered by automated tests and deployable to another org. Run as **Admin Tester**.

### E2E-45 Full unit-test run at ≥75% coverage

US-14 AC part 1 · NFR-03 · Must

#### STEPS

- Terminal: `sf apex run test --code-coverage --result-format human --wait 10 -o <org>`

#### EXPECTED

✓ **All tests pass** (reference build: 37/37) and overall coverage ≥ **75%**. Record: passed \_\_\_\_/\_\_\_\_ · coverage \_\_\_\_%.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_

### E2E-46 Deployment validates against a clean org

US-14 AC part 2 · NFR-06 · Must

#### STEPS

- Terminal: `sf project deploy start --source-dir force-app -o <secondOrg> -l RunLocalTests --dry-run`

#### EXPECTED

✓ The validate-only run **succeeds** with all tests passing — the release is deployable on demand.

Result: Pass ☐ Fail ☐ N/A ☐ · Tester: \_\_\_\_\_ · Date: \_\_\_\_\_ · Defect #: \_\_\_\_\_



## 15 Traceability matrix & sign-off

Every acceptance criterion, mapped to the case(s) that prove it. Exit criteria (§1.5) require a Pass in every row.

### 15.1 Traceability — core user stories

| STORY         | ACCEPTANCE CRITERION (ABRIDGED)                                     | PROVEN BY                           | PASS?                    |
|---------------|---------------------------------------------------------------------|-------------------------------------|--------------------------|
| US-01         | Catalogue objects, cohort belongs to course, cohort attributes      | E2E-01, E2E-02                      | <input type="checkbox"/> |
| US-02         | Enrolment links Contact+Cohort; date/status defaults                | E2E-05                              | <input type="checkbox"/> |
| US-03         | Enrolled Count (excl. cancelled) & Seats Remaining, no manual entry | E2E-03, E2E-04                      | <input type="checkbox"/> |
| US-04         | Overbooking blocked with clear error; holds at 200 records          | E2E-06, E2E-07, E2E-18              | <input type="checkbox"/> |
| US-05         | Duplicate enrolment rejected                                        | E2E-08, E2E-18                      | <input type="checkbox"/> |
| US-06         | Confirmation email + follow-up task                                 | E2E-10                              | <input type="checkbox"/> |
| US-07         | Cohort auto-Full at zero seats                                      | E2E-11                              | <input type="checkbox"/> |
| US-08         | Guided wizard; error if full                                        | E2E-12, E2E-13                      | <input type="checkbox"/> |
| US-09         | Reminder T-3 days, confirmed only                                   | E2E-14                              | <input type="checkbox"/> |
| US-10         | Console: list, filter, seats, enrol, toast                          | E2E-16, E2E-17                      | <input type="checkbox"/> |
| US-11 / US-12 | Agent answers availability / status from live data                  | E2E-19, E2E-20 (else N/A + Suite L) | <input type="checkbox"/> |
| US-13         | Contact Total Enrolments auto-updated                               | E2E-15                              | <input type="checkbox"/> |
| US-14         | 75%+ tested; deployable to another org                              | E2E-45, E2E-46                      | <input type="checkbox"/> |
| NFR-01/02     | Bulk-safe, no SOQL/DML in loops                                     | E2E-07 (behavioural) + code review  | <input type="checkbox"/> |
| NFR-04        | Human-readable errors                                               | E2E-06, E2E-13, E2E-18              | <input type="checkbox"/> |

### 15.2 Traceability — extension features

| CRITERION                                                                                                                    | PROVEN BY           | PASS?                    |
|------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------|
| EXT-INV-1...4 — auto-invoice at fee/+14d · idempotent · free courses excluded · re-issue after cancel · Paid needs reference | E2E-22...26         | <input type="checkbox"/> |
| EXT-PAY-1...4 — settle with reference & roll-up · declined unchanged · no double pay · audit complete                        | E2E-27...30, E2E-42 | <input type="checkbox"/> |
| EXT-CERT-1...3 — issue on completion · once only · correct PDF content                                                       | E2E-31...33, E2E-43 | <input type="checkbox"/> |
| EXT-CCY / I18N — configured rates · frozen at issue · French UI & certificate                                                | E2E-34...36         | <input type="checkbox"/> |
| EXT-ERR-1...3 — Apex, flow, and rollback-surviving logs                                                                      | E2E-37...39         | <input type="checkbox"/> |
| EXT-POR-1...5 — login · privacy · pay · certificate · access control                                                         | E2E-40...44         | <input type="checkbox"/> |

### 15.3 Execution summary

| METRIC | COUNT |
|--------|-------|
|--------|-------|

|                                                 |                           |
|-------------------------------------------------|---------------------------|
| Cases executed / total                          | _____ / 46                |
| Passed · Failed · N/A                           | _____ · _____ · _____     |
| Defects raised (Critical / High / Medium / Low) | ____ / ____ / ____ / ____ |
| Unit test result & coverage (from E2E-45)       | _____ passed · _____ %    |

## 15.4 Sign-off

**Test outcome:** Accepted ☐ Accepted with conditions ☐ Rejected ☐

Conditions / notes: \_\_\_\_\_

**Lead tester:** \_\_\_\_\_ **Signature:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Product owner / client:** \_\_\_\_\_ **Signature:** \_\_\_\_\_ **Date:** \_\_\_\_\_